



# New insights into the technical challenges of the Mauritius long term energy strategy



M.N. Edoo\*, Robert T.F. Ah King

Department of Electrical and Electronic Engineering, Faculty of Engineering, University of Mauritius, Réduit, 80837, Mauritius

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## ABSTRACT

The Mauritius Long Term Energy Strategy targets 35% of electricity generation from renewables by 2025. This work presents the modeling of the Mauritian energy system in the EnergyPLAN software to identify and gain insights into the major technical challenges involved to achieve that target. Based on simulation of alternative scenarios, it is shown that if solar PV is to be the main source of renewable energy by 2025, operation of the conventional generation fleet at very low levels (20–30% of instantaneous demand) and high ramping requirements become indispensable to avoid significant curtailment and meet the net load. Alternatively, the use of AC-coupled battery energy storage for curtailment reduction was found to be economically unviable due to low utilization. Curtailment to increase solar PV penetration was found to be effective as allowing a modest curtailment rate increased penetration significantly. Consequently, new policy implications are derived. First, it is vital that the generation expansion planning methodology prioritize generation flexibility requirements. Second, the techno-economic feasibility of using DC-coupled Solar plus storage should be investigated as it circumvents the grid stability problem. Third, an effective curtailment policy should be formulated to optimize the contribution of solar PV in the national electricity mix.

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## 1. Introduction

The Republic of Mauritius consists of several islands located in the Indian Ocean with the main island of Mauritius located about 2000 km off the south-east coast of Africa. The main island of Mauritius (hereafter Mauritius) is home to a population of 1.3 million people which is around 97% of the total population [1]. Like many Small Island Developing States, the country is heavily dependent on imported fossil fuels for power generation, transport and industry. For year 2016, imported fossil fuels accounted for

around 85% of the nation's total primary energy requirement. The corresponding cost was around USD 602.9 million (MUR 21.1 Billion<sup>1</sup>) which represented 12.8% of the nation's total import bill [2]. Undoubtedly, the Mauritian authorities have devised strategies to reduce the dependence on fossil fuels. One fundamental way is to harness the local renewable energy (RE) Mauritius is endowed with. Consequently, like in many other countries, the electricity sector has been the prime target of decarbonization efforts since there exists several mature technologies that can cost-effectively harness the RE sources.

### 1.1. Present situation of the energy sector in Mauritius

The total primary energy requirement of Mauritius in 2016 was 1550 kilotonne of oil equivalent (ktoe) out of which 85% was met from imported fossil fuels and the rest 15% was from local renewable sources [2]. The breakdown of the different energy sources is as shown in Fig. 1.

**Abbreviations:** AC, Alternating current; BESS, Battery Energy Storage System; CEB, Central Electricity Board; DC, Direct Current; GWh, gigawatt-hour; ktoe: kilotonne of oil equivalent, LCOS: Levelised cost of storage, LTES: Long Term Energy Strategy, MARENA: Mauritius Renewable Energy Agency, MGSS: Minimum grid stabilisation share, MUR: Mauritian rupee, NSG: Non-synchronous generation, OCGT: Open cycle gas turbine, p.u: per unit; PV, Photovoltaic; RE, Renewable energy; USD, United States dollar; VRE, Variable renewable energy; WTE, Waste-to-energy.

\* Corresponding author.

E-mail address: [nawazedoo@gmail.com](mailto:nawazedoo@gmail.com) (M.N. Edoo).

<sup>1</sup> 1 United States Dollar (USD) = 35 Mauritian rupees (MUR).

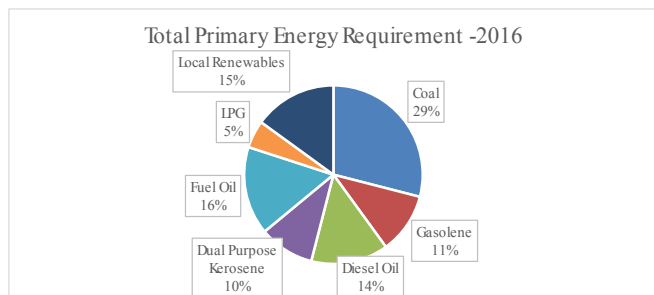


Fig. 1. Total primary energy requirement 2016 [2].

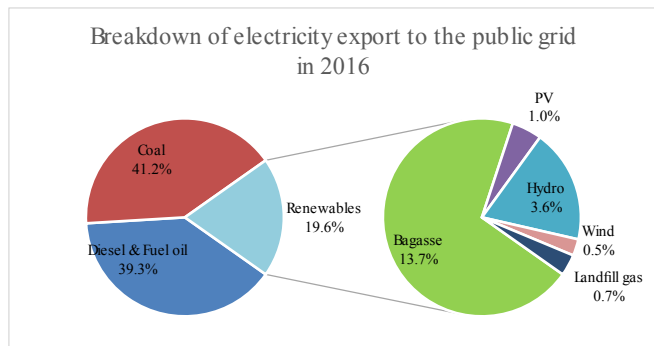


Fig. 2. Breakdown of electricity export to grid (adapted from Ref. [2]).

The contribution of renewables is mainly seen in the electricity sector where the peak power and electrical energy demand for 2016 was 467.9 MW and 2737.6 GWh, respectively. As per the Statistics office, in 2016, the share of renewable energies in the electricity mix stood at 21.8% [2]. This percentage includes the electricity derived from bagasse<sup>2</sup> and used internally in private sugar mills. However, in this analysis, for the determination of the annual penetration of renewable energy, only the electricity that is exported to the public electricity grid shall be considered so as to be in line with the Long-Term Energy Strategy 2009–2025 (LTES)<sup>3</sup> [4]. Consequently, as per this approach, the share of renewable electricity is estimated to be 19.6%.<sup>4</sup> The breakdown is shown in Fig. 2.

## 1.2. Long-term energy strategy 2009–2025

The LTES target of 35% RE in the electricity mix by 2025 was expected to be met progressively as shown in Table 1. From 2010 to 2015, the country underperformed on its objectives mainly because the expectations on Waste-to-Energy (WTE) and to a lesser extent, wind, failed to materialize. Protests by the general public due to fear of pollution led to the cancellation of a 20 MW WTE project in 2009 [5]. For 2020, as it will be demonstrated later, the share of RE is expected to reach around 25.4% in the best-case scenario.

<sup>2</sup> The dry pulpy residue left after the extraction of juice from sugar cane. It is used as fuel for power generation.

<sup>3</sup> To demonstrate that the LTES considered the electricity export to the grid to determine the share of RE in the island's electricity mix, note that the baseline contribution of bagasse in the LTES is stated to be 350 GWh in 2009 in contrast to the total production of bagasse-based electricity which was 485 GWh in that year [3]. Thus, there can be no doubt regarding which figure to use.

<sup>4</sup> In the absence of publicly available data concerning export of bagasse-based electricity for calendar year 2016 at the time of writing, it is reasonably assumed that the contribution of bagasse for year 2016 is 376 GWh which is approximately equal to year 2015.

Since its publication in 2009, the LTES has developed many inconsistencies. For example, it targeted 8% of electricity generation from wind power in 2025 and only 2% from solar photovoltaic (PV). The reverse situation appears to be unfolding due to the dramatic drop in price of solar PV and the difficulties in obtaining development permits for wind power plants locally. Clean coal technology is also no longer pursued as the authorities now pursue less-polluting options [8]. Furthermore, the prospect of harnessing geothermal energy is extremely remote [9].

As for the prospect of increasing electricity generation from bagasse, this may appear uncertain due to the rapidly decreasing cultivation area [10]. Nonetheless, the LTES recommended that new bagasse-fired plants be equipped with higher pressure boilers so that more electricity can be generated with same amount of bagasse (from exports of 350 GWh to 600 GWh). Finally, WTE is not delivering on the LTES target as it provides only 0.6% instead of the expected 5% of electricity needs [2].

To correct those inconsistencies and chart a new strategy, the Ministry of Energy and Public Utilities very recently presented a RE Roadmap to 2030 which confirms the level of RE at 35% for 2025 and maintains it to 2030 [11]. The Roadmap essentially swaps the contribution of wind and solar in recognition of the greater role the latter technology will play and removes geothermal power in the future generation mix. However, the Roadmap still lays as much emphasis on increased contribution of bagasse and WTE as the LTES despite the difficulties in the sugar industry and the lack of progress in WTE. It is evident therefore that uncertainties abound concerning the way to achieve the target of 35% green electricity. To be able to capture all the significant implications, a scenario approach coupled with modelling and simulation is required.

It is also instructive to briefly compare the Mauritian electricity sector and RE targets with other islands that share similar characteristics. Thus, two islands were chosen namely the French island of Réunion which offers an excellent case study as it is very similar in many aspects as well as being geographically close, and the US Hawaiian island of Oahu. The pertinent statistics for each island are shown in Table 2.

Réunion island targets complete decarbonization of its electricity generation mix by 2030 [12,13]. For year 2018, the solar PV installed capacity was 190 MW<sub>DC</sub> and since 2013 the installation of new PV capacity has significantly slowed to essentially come to a halt in 2015 even though the targets for 2018 was 239.9 MW<sub>DC</sub> and 307.9 MW<sub>DC</sub> in 2023 [13]. An official report [14] identified several constraints chief among which is the limit of 30% instantaneous penetration (this has been now increased to 35%) which occasionally requires curtailment of variable renewable energy (VRE) generation without compensation. As the frequency and duration of the curtailments are not known in advance and the contracts inform the possibility of disconnection 100% of time, the bankability of future projects is affected (it is unclear if the policy has changed since publication of [14]). It is noteworthy that the latest curtailment rate in La Réunion is very low at 0.05% [13]. Arguably, the consequence of the policy has been to constrain the growth of VRE capacity to strictly limit the amount of excess energy to almost zero. This then raises an interesting question on what the acceptable level of curtailment is to balance the growth of PV generation and the rate-payers' interest. Apart from that, Réunion island leads the comparison in terms of share of RE in the electricity mix thanks in large measure to the substantial hydropower resource. In contrast, Mauritius has a much smaller hydro potential which has been almost completely tapped [15]. The source of data for La Réunion is from Ref. [13].

The island of Oahu Hawaii targets 100% of its electricity from RE by 2045 [16]. Even though it is the smallest of the islands, it has the highest consumption of electricity (sales only shown in Table 2)

**Table 1**  
LTES Targets and Actual Performance.

| Fuel Source   |            | Percentage of Total Electricity |            |        |            |        |        |
|---------------|------------|---------------------------------|------------|--------|------------|--------|--------|
|               |            | 2010                            |            | 2015   |            | 2020   | 2025   |
|               |            | Target                          | Actual [6] | Target | Actual [7] | Target | Target |
| Renewable     | Bagasse    | 16%                             | 14.7%      | 13%    | 14.2%      | 14%    | 17%    |
|               | Hydro      | 4%                              | 4.2%       | 3%     | 4.5%       | 3%     | 2%     |
|               | WTE        | 0                               | 0          | 5%     | 0.8%       | 4%     | 4%     |
|               | Wind       | 0                               | 0          | 2%     | 0.0%       | 6%     | 8%     |
|               | Solar PV   | 0                               | 0          | 1%     | 0.8%       | 1%     | 2%     |
|               | Geothermal | 0                               | 0          | 0      | 0.0%       | 0      | 2%     |
|               | Sub-total  | 20%                             | 18.9%      | 24%    | 20.3%      | 28%    | 35%    |
| Non-Renewable | Fuel Oil   | 37%                             | 39.7%      | 31%    | 40.7%      | 28%    | 25%    |
|               | Coal       | 43%                             | 41.3%      | 45%    | 38.9%      | 44%    | 40%    |
|               | Sub-total  | 80%                             | 81.1%      | 76%    | 79.7%      | 72%    | 65%    |
|               | Total      | 100%                            | 100.0%     | 100%   | 100.0%     | 100%   | 100%   |

**Table 2**  
Comparison of some island power systems.

|                                 | Mauritius | Réunion | Oahu             |
|---------------------------------|-----------|---------|------------------|
| Year                            | 2016      | 2018    | 2018             |
| Area (km <sup>2</sup> )         | 1864 [18] | 2504    | 1545 [19]        |
| Electrical energy demand (GWh)  | 2737.6    | 2958.9  | 6549 [20]        |
| % Renewable                     | 19.6%     | 36.5%   | 22%              |
| % VRE                           | 1.5%      | 9.0%    | 15%              |
| % Solar PV                      | 1.0%      | 8.6%    | 12%              |
| % Wind                          | 0.5%      | 0.4%    | 3%               |
| % Hydro                         | 3.6%      | 20.3%   | 0.0%             |
| % Biomass, Biogas, Biofuel etc. | 13.7%     | 7.1%    | 1%               |
| % WTE                           | 0.7%      | 0.0%    | 6%               |
| % curtailment                   | 0.0%      | 0.05%   | 0.46%–2.05% [16] |

which indicates a very affluent and energy intensive society. Yet, it also has the highest share of VRE in the electricity mix which indicates that considerable effort is being made to transition to a more sustainable energy system. Oahu also has a very large proportion of solar PV electricity coming from rooftop installations (10% versus 2% for grid-scale solar) which is a wise strategy given that on small islands, land area is limited. The curtailment rate in Oahu is 2.05% relative to curtailable VRE and 0.46% relative to all RE including firm RE [16]. This is to be contrasted with La Reunion's case. Note that the curtailment rate includes curtailment due all possible reasons including system constraints. The source of data for Oahu is [17] unless specified otherwise.

### 1.3. Modelling of the Mauritian energy system

The number of studies devoted to the modelling of the Mauritian energy system is quite small. Most of the studies focus on the electricity part of the energy system. Balnac et al. [21] develop a system dynamic model for the electrical power system with the objective of forecasting electric energy demand. Welsch et al. [22] model the energy system and focus on presenting a decision making and policy approach based on Climate, Energy, Water and Land-Use Systems. It therefore does not focus on the transition to large-scale renewables integration. A report by a private company [23] concluded that Mauritius is on track to achieve the 35% target of RE in the electricity generation by 2025 and can achieve a 60% penetration by 2035. However, limitations of the study are that for 2025, the study is very optimistic concerning the share of wind energy at 8% whereas wind is producing only 0.5% (2016) and individual projects generally face significant clearance issues for implementation. The share of WTE at 5% is also overly optimistic given the present share of 0.6% [2]. Furthermore, the capacity factor assumed for wind generation is 30.2% which is very high given that

historical records imply that capacity factor in Mauritius is around 17.7% [2]. Finally, the study does not consider the integration of the transport sector as it is outside its scope. In contrast, Khoodaruth et al. [10] review the technologies that can usher Mauritius into an era of a fully renewable energy system by 2050 but do not model the energy system to carry out a techno-economic study. This deficiency is also present in the official LTES.

In a very recent study, Timmons et al. [24] performed an optimization for a 100% renewable electricity system for Mauritius for year 2040 using the OSeMOSYS software. The main finding is that the minimum-cost RE portfolio consists of approximately equal proportions of solar, wind, and biomass electricity, along with electricity storage. Additionally, several policy issues regarding RE targets, policy instruments to foster RE growth and land use among others are discussed. The study focuses on the long-term policy aspect of the energy transition and does not delve in the short-to-medium term technical challenges that must first be overcome before a 100% RE system can be contemplated. This may be because the model used a relatively low temporal resolution with only two time slices per day to reduce computational burden. Finally, in the Roadmap 2030 [11], the authorities employed load duration curves of the projected national electricity demand to make a techno-economic assessment of introducing new generation capacity including the VREs. It can be argued that, in this case, the absence of simulation at sufficiently high temporal resolution (at least hourly) precluded the identification of technical challenges that lie in the path of the 35% target and beyond.

The Mauritius Renewable Energy Authority (MARENA) [25] and the World Bank [26] recognize the importance of adequate modeling tools specially to capture the temporal availability of VREs. The absence of appropriate models for the Mauritian energy system is thus brought into sharp focus as a clear research gap. Having an appropriate model of the energy system will enable better discussions on the future of the system. More importantly, this could prevent investment pitfalls that can potentially lock the energy system in an inflexible configuration that could prevent evolution towards large-scale penetration of renewables in the long term.

### 1.4. Instantaneous non-synchronous generation limit in island power system

To manage the integration of VRE, a few isolated island power systems have established instantaneous non-synchronous generation (NSG) limits for VRE generation in the power system. The instantaneous NSG is defined as:

$$\%NSG = \frac{\text{Total Non Synchronous Generation}}{\text{Total Generation}} \times 100\% \quad (1)$$

Accordingly, in the LTES [4], the maximum wind capacity that can be installed is stated to be 30% of the night load implying that NSG limit for the island of Mauritius is 30% [4]. In contrast, the Integrated Electricity Plan 2013–2022 prepared by the national electricity utility, the Central Electricity Board (CEB) states that the maximum amount of VRE that can be safely integrated in an insular system is about 15–20% of the prevailing load demand [15]. It is to be noted however that Mauritius has up to now not set any official NSG limit with regards to system operation.<sup>5</sup>

### 1.5. Curtailment of variable renewable energy

For the island of Mauritius, the frequency stability both in steady state and dynamic time frames are stated to be the limiting factors for penetration of renewables [4,15,25,28]. In contrast, the potential issue of excess electricity production from the VRE sources has not been explicitly mentioned as a future challenge in the official documentation.

Once considered as an unthinkable economic burden, curtailment of variable renewables could now become a rational economic decision for maximizing VRE penetration and lowering overall system costs [29]. Consequently, it would be pertinent to investigate the potential issue of excess electricity generation arising from large scale VRE in anticipation of capacity increases.

### 1.6. Energy storage for time-shifting of renewable energy

In Mauritius, the power-intensive application of Battery Energy Storage Systems (BESS) in the form of frequency regulation is being adopted [28]. However, it is now becoming apparent that energy storage with longer duration will play a key role to integrate large amounts of VRE into the grid, specifically to reduce curtailment/time-shift by charging during periods of excessive RE generation and discharging when there is sufficient demand [30–33]. Anticipating the difficulties of the conventional generation fleet to provide the required flexibility in the short term, it becomes important to study how BESS for energy time-shift can help in achieving the 35% target.

### 1.7. Scope of this work

The scope of this work is two-fold. First, the objective is to develop a model of the Mauritian energy system adequate for the modelling of large-scale integration of variable renewables.

Second, at this crucial juncture where the energy system is undergoing profound changes, it becomes necessary to analyze the country's long-term strategy. Consequently, another objective of this work is to model the alternative pathways by which Mauritius may achieve the LTES target and thereby identify and gain valuable insight concerning potential technical challenges to further deployment of VRE. Strategies and research avenues to overcome those issues will then be formulated.

## 2. Data and methods

### 2.1. Energy-system analysis tool

The EnergyPLAN energy-system analysis tool has been chosen for this study. It is an input/output deterministic energy system analysis model that aims to identify optimal energy system designs and operation strategies using hourly simulations over a 1-year time period [34]. Given that there is a strong likelihood that Mauritius will further expand its VRE capacity base, the ability to perform hourly simulations has been a key advantage for EnergyPLAN. The fact that the model considers the energy system in aggregate is also an important advantage given that detailed data especially concerning the power generation plant is not publicly available. The ability of EnergyPLAN to model not only the electricity sector but also the transport and industrial subsectors was also a deciding factor in its selection. This is because establishing synergies between the electricity and transport sector will be an important element in the decarbonization of the whole energy sector locally in the long term [34].

The EnergyPLAN tool, which is a freeware, has been used in many studies that cover a wide range of energy planning issues and is therefore a recognized analysis tool. For example, it is used for the analysis of the flexibility requirements of the power system for large scale integration of wind energy in China's Jiangsu province [35], the analysis of contemporary energy systems for, determination of maximum feasible wind integration in Ireland [36], investigating the benefits of energy storage and/or interconnection in Great Britain [30] and analyzing the impacts of integrating VRE in long-term future scenarios of Colombia [37]. EnergyPLAN has also been extensively used for the analysis of national 100% renewable energy systems - Denmark [38], Ireland [39], Croatia [40], Macedonia [41], the Åland islands [42], the island of Gran Canaria [43] and more sustainable energy systems for cities as well [35,44]. More details on the EnergyPLAN tool can be found at [45].

### 2.2. Energy demand forecast

The World Bank's Sensitized Low Scenario Electricity Sent-out Forecast [26] was used as input for the yearly electricity demand for 2020 i.e. 2977.5 GWh. Given that the forecast ends with 2022, it was extended up to 2025 with a constant yearly growth factor of 1.21%. Thus, the electricity demand for 2025 is projected to be 3163.1 GWh.

### 2.3. The reference energy model year 2016

The building of a reference energy model is the first step in the modelling exercise. The Reference Model is based on Year 2016 given that this was the latest year for which a complete data set was available when the modelling works began. The simulation results of the Reference model are compared against actual statistics of Year 2016. The Reference Model is deemed validated if the difference between simulation and reality is small enough.

Data for the model was obtained from the Annual Energy & Water Report 2016 prepared by Statistics Mauritius [2] and from CEB. Statistics Mauritius is the official organization responsible for collection, compilation, analysis and dissemination of the official data of Mauritius. Table 3 shows the specific data used and their sources. Note the model requires the conversion efficiency of the large thermal power plant system and, from data in Ref. [2], this was determined to be 29.9% in 2016. This relatively low efficiency is due to coal-bagasse Independent Power Producer plants which have conversion efficiencies of 25.0% and 23.6% for coal and bagasse respectively. This is to be contrasted with the CEB's plants which

<sup>5</sup> The approach is different in the neighboring Island of Reunion where the authorities have set an official limit of 35% [13] while in Corse the French Authorities have set a limit of 35% in 2018 which they plan to increase to 45% in 2023 [27].



**Table 3**  
Input to the reference model.

| Input                                                                                            | Reference 2016 | Source                 |
|--------------------------------------------------------------------------------------------------|----------------|------------------------|
| Electricity demand (GWh)                                                                         | 2737.6         | [2]                    |
| Hydropower generation (GWh)                                                                      | 99.5           |                        |
| Landfill Gas (GWh)                                                                               | 19             | [47]                   |
| Power generation capacity                                                                        |                |                        |
| Solar PV (MW)                                                                                    | 16.2           |                        |
| Wind (MW)                                                                                        | 9.35           |                        |
| Hydro (MW)                                                                                       | 56.3           |                        |
| Effective Thermal Generation capacity (MW)                                                       | 594            | [4,15]                 |
| Non-synchronous generation limit                                                                 | 15%–30%        |                        |
| Efficiency of thermal generation                                                                 | 29.9%          | Based on [2]           |
| Transport demand                                                                                 |                | [2]                    |
| Jet Fuel (GWh)                                                                                   | 1716.3         |                        |
| Diesel (GWh)                                                                                     | 1993.9         |                        |
| Petrol (GWh)                                                                                     | 2125.8         |                        |
| LPG (GWh)                                                                                        | 43.7           |                        |
| Industry, Commercial and Residential fuel demand (incl. sugar mill assumed internal consumption) |                | [2]                    |
| Oil (GWh)                                                                                        | 851.4          |                        |
| Ngas including Liquid Petroleum Gas (GWh)                                                        | 897.8          | Adapted from Ref. [2]. |
| Bagasse including biomass (GWh)                                                                  | 872.9          |                        |
| Coal (GWh)                                                                                       | 784.5          |                        |

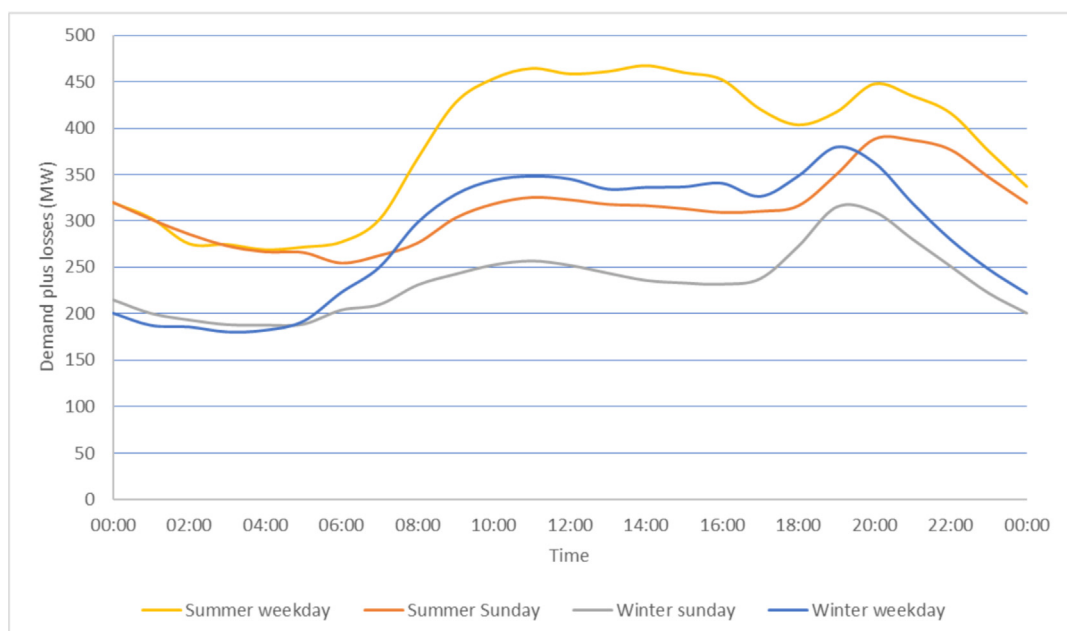
have an overall efficiency of 44.5% reference [46]. It is clear therefore that there is a huge potential for improving the efficiency of the private plants, reduce the primary energy consumption of the island and the import bill of fossil fuels.

All hourly profiles were obtained from the electric utility, the CEB. The solar PV and wind generation profiles are based on the actual output of one large solar PV plant and wind farm respectively for the whole year 2016. Thus, the VRE generation profiles captures the daily and seasonal variations at an acceptable resolution of 1 h. Figs. 3 and 4 below show the profiles for demand and solar generation for a few pertinent days.

Concerning the demand profiles in Fig. 3, it can be seen that there is significant difference between the weekdays and weekends (Sunday shown as an example) on one hand and the summer and winter days on the other. The difference between summer and winter demand, especially during weekdays is huge. For the

example shown, the daytime demand during summer is 28% more than in winter. This is mainly due to the proliferation of air conditioning load over the whole island.

Fig. 4 shows PV generation profiles under different conditions. Of particular interest is the variability experienced during cloudy days. For example, in the summer cloudy day example, from 10:00 to 11:00, the power output drops from 0.75 per unit of rated capacity (p.u) to 0.35 p.u but increases to 0.59 p.u in the next hour. As this profile is that of a single plant, albeit a large utility-scale one, the p.u variations are expected to be somewhat larger than the whole PV system distributed over the island. Nonetheless, bearing in mind that the longest straight-line distance on the island is only around 60 km, a large cloud front could affect a large part of the island in a relatively short time frame. Consequently, as a worst-case scenario and in the absence of more detailed data, it is justifiable that the variable RE system be modelled by scaling up a single



**Fig. 3.** Selected daily demand profiles.

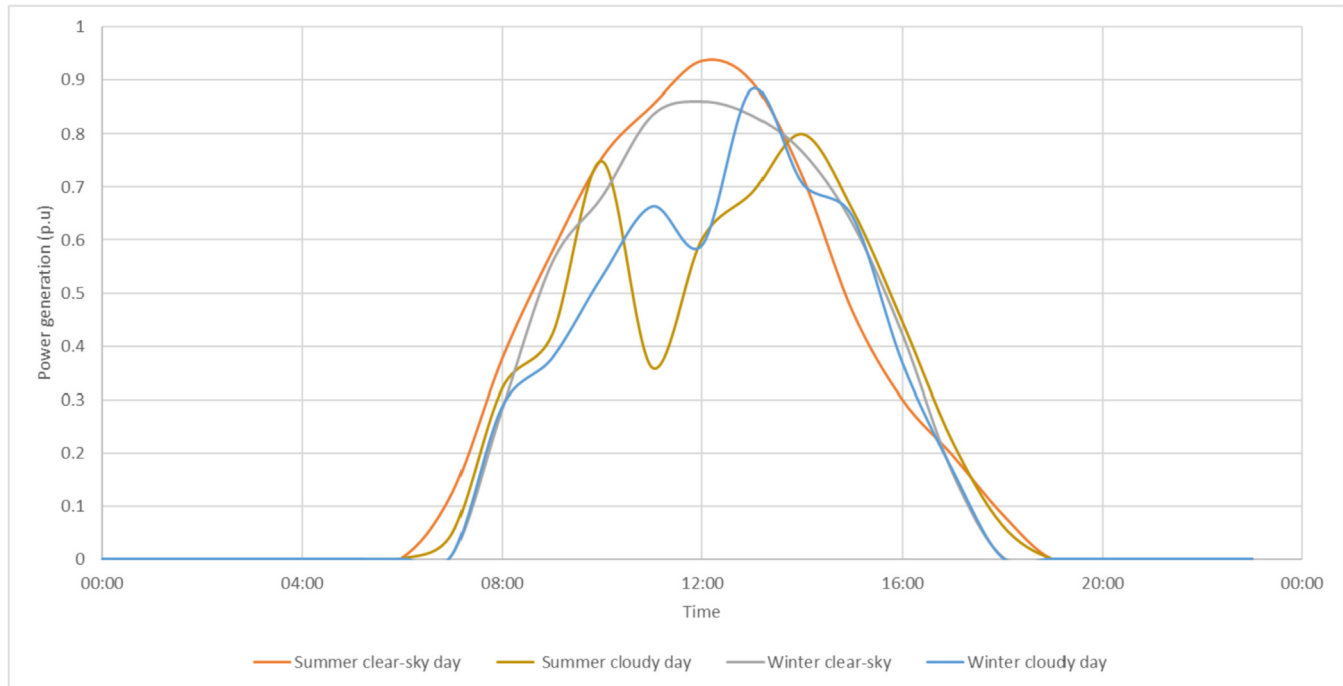


Fig. 4. Solar PV generation profiles for a few specific days.

large plant. It is to be noted that the technical simulation is being employed as opposed to the market-economic simulation because the Mauritian electricity sector is regulated in contrast to the market-based system in developed countries.

## 2.4. Scenarios

The Scenarios to be studied will be based on the interim year 2020 and final year 2025. Given that at the time of writing there are no quantifiable targets for reducing fossil energy use by electrification in the transport and industrial sectors up to 2025, the scenarios will focus on the electricity sector. Further works will include the whole energy sector especially transportation to study how to achieve large scale integration of RE.

### 2.4.1. Scenario 2020

For Scenario 2020, the Reference Energy Model power plant capacity will be updated as follows:

- Power plant capacity increased to reflect addition of a new Open Cycle Gas Turbine (OCGT) power plant of capacity  $2 \times 40$  MW [48].
- Solar PV capacity increased to 116 MW<sub>ac</sub> to reflect the committed projects.
- Wind capacity increased to 38.75 MW to reflect coming into operation of the new plant.
- The electricity demand is increased to 2977.5 GWh in line with the World Bank's forecast.

The list of committed projects was determined based from the list of Environment Impact Assessments on the Ministry of Environment's website [49] and it is assumed they will be commissioned by 2020.

### 2.4.2. Scenarios 2025

Despite the official target of the LTES and the contribution

specified for each type of RE, one cannot rely on the LTES due to its multiple limitations. Therefore, four scenarios will be considered namely LTES, PV, PV plus Storage and Curtailment. In all the scenarios, the electricity demand was set to 3163.1 GWh.

### 2.4.3. LTES scenario

This is broadly informed by the official Strategy as per the following:

- Contribution of electricity from bagasse will increase to provide 17% of the total demand i.e. approx. 540 GWh annually.
- Contribution from Waste to Energy plants to increase to provide 4% of the total demand i.e. approx. 125 GWh annually.
- Wind capacity is maintained at the 2020 level i.e. 38.75 MW.
- Solar PV capacity is increased to 163 MW<sub>ac</sub>.
- The Combined Cycle Gas Turbine power plant is completed and total capacity is 120 MW.
- Hydro contribution remains unchanged.

This scenario also captures the gist of the recent RE Roadmap 2030 in that it relies heavily on increased contribution from bagasse and WTE.

### 2.4.4. PV scenario

This Scenario aims to capture the eventuality that the country turns massively to PV generation to achieve its 35% target. This is a credible scenario as most VRE capacity additions are solar PV and additional significant contributions from bagasse and WTE are not certain. The following points are to be noted:

- The contribution of bagasse and hydro will stay at their present levels i.e. a total of around 475 GWh annually.
- In view of the difficulties encountered by wind generation project locally, the wind capacity is increased modestly by 20 MW over the 2020 level up to 58.75 MW.

- The rest of the required RE will be modelled to come mainly from PV. Thus, the PV installed capacity will be increased to 322 MW<sub>ac</sub> to make up for all the balance of energy required to reach the target. This figure is based on the available energy that if fully integrated in the system would enable the country to reach its target of 35% RE.

#### 2.4.5. PV plus storage scenario

The PV plus Storage Scenario is similar to the PV scenario but with the difference that large scale lithium-ion BESS designed for energy-time shift (curtailment reduction) of excess RE is considered. In contrast to the previous scenarios where the NSG limit is varied, in this scenario it will be assumed that by year 2025, the necessary grid modernization has been performed so that an NSG limit of 50% is achieved. It is felt that such a level can realistically be achieved by 2025<sup>6</sup> and that the authorities will resort to BESS for curtailment reduction only when all generation flexibility options have been exhausted i.e. not at NSG limits lesser than 50%. The round-trip efficiency of the BESS is assumed to be 85% and the simulations are performed for varying MW capacity of BESS starting from 20 MW in increments of 20 MW up to a maximum of 180 MW. Storage durations of 2 and 4 h have been chosen.

#### 2.4.6. Curtailment scenario

In the Curtailment Scenario, the solar PV installed capacity will be increased from the base value of 100 MW up to 300 MW and the amount of excess energy versus accepted energy will be analyzed. The rest of the system will be as in the PV Scenario. The NSG limit for this scenario was set at 50% in line with realistic expectations.

#### 2.5. Stability constraints

In EnergyPLAN, the corresponding parameter which specifies the NSG limit is the Minimum Grid Stabilisation Share (MGSS) which specifies limitations on the operation in order to assure grid stability [51]. The MGSS can be expressed in terms of the NSG Limit as follows:

$$MGSS = 100\% - NSG \text{ Limit} \quad (2)$$

The simulations for Scenarios 2020 and 2025 LTES and PV shall be performed for a range of NSG limits from 20% to 100% (MGSS of 80%–0%) and the 100% being a fully inertia-less grid which is not expected to become reality by 2025 but will provide an interesting case.

### 3. Results and discussion

#### 3.1. Validation of the EnergyPLAN model

After all the required parameters have been input, the Reference Energy Model 2016 is simulated. For 2016, the electricity demand was 2737.6 GWh and based on the normalized hourly demand profile, the model calculates the actual power demand. Thus, a key test of the model accuracy is the comparison of the simulated power demand at specific hours and the peak power demand with real life. Table 4 below shows how the simulated values of power demand compare with reality. Furthermore, the peak demand of 2016 was simulated at 468.0 MW compared to actual 467.9 MW [2]. The simulated values are very close to the real values.

**Table 4**

Comparison of average real and simulated power demand for Reference Energy Model 2016.

| Month                      | Real/MW | Simulated/MW | % Difference |
|----------------------------|---------|--------------|--------------|
| January                    | 347.3   | 347.4        | −0.03        |
| February                   | 347.8   | 347.8        | 0.00         |
| March                      | 351.2   | 351.2        | 0.00         |
| April                      | 335.6   | 335.6        | 0.00         |
| May                        | 301.7   | 301.7        | 0.00         |
| June                       | 281.8   | 281.9        | −0.04        |
| July                       | 270.7   | 270.8        | −0.04        |
| August                     | 275.3   | 275.4        | −0.04        |
| September                  | 276.0   | 276.1        | −0.04        |
| October                    | 299.1   | 299.1        | 0.00         |
| November                   | 317.9   | 318.0        | −0.03        |
| December                   | 336.1   | 336.1        | 0.00         |
| Peak Power Demand for 2016 | 467.9   | 468.0        | −0.02        |

**Table 5**

Comparison of real and simulated generation from RE sources for Reference Energy Model 2016.

| Source            | Real/GWh [2] | Simulated/GWh | Difference (%) |
|-------------------|--------------|---------------|----------------|
| Photovoltaic      | 26.4         | 26.3          | 0.38           |
| Wind              | 14.5         | 14.3          | 1.38           |
| Biomass (Bagasse) | 375.9        | 378.2         | −0.56          |
| Thermal           | 2202.8       | 2200.1        | 0.13           |
| Hydro             | 99.5         | 99.5          | 0.0            |
| Landfill gas      | 19.0         | 19.0          | 0.0            |

**Table 6**

Comparison of real and simulated values of fuel consumption.

| Source                      | Real/GWh [2] | Simulated/GWh | Difference (%) |
|-----------------------------|--------------|---------------|----------------|
| Petroleum products          | 10,096.2     | 10,110.5      | −0.14          |
| Coal                        | 5294.6       | 5323.2        | −0.54          |
| Biomass (including bagasse) | 2470.8       | 2475.3        | −0.18          |

The representations of variable renewable and thermal energy are also checked as shown in Table 5. It is to be noted that the simulation for the Reference Energy System was performed at NSG Limits of 15%, 20% and 30% as per the limits set in Refs. [4,15]. In all these simulations, there was no curtailed renewable energy which agrees with real life as there have been no official reports of curtailment due to system-wide excess of VRE.

The amount of fuel used versus EnergyPLAN simulation is shown in Table 6.

For the whole energy system, the real and simulated values of Total Primary Energy Requirements as well as the different fuels are compared. Clearly, the model simulation results are in very close agreement with reality. Consequently, the EnergyPLAN model for Mauritius is deemed validated and can be used for simulation of advanced energy system for the island.

#### 3.2. Scenario 2020

The variation of excess solar PV energy with increasing NSG levels is shown in Fig. 5. As it can be seen, operation of the Mauritian power system in 2020 with NSG limit of at least 40% ensures that virtually all the VRE available is integrated (excess solar energy of 0.43 GWh or 0.2% of total available solar PV energy). In contrast, limiting the instantaneous penetration of VRE to 30% as previously in neighboring Reunion island could result in excess electricity production of around 4.63 GWh or 2.4% of total solar PV energy available. The case of NSG of 20% is shown to be extremely conservative and totally unacceptable. The NSG limit of 40% is

<sup>6</sup> The French authorities have targeted an NSG of 45% by 2023 for the island of Corse [27] and, in the case of the island of Oahu, experts have acknowledged that 50% is the point after which more grid stability studies are needed [50].

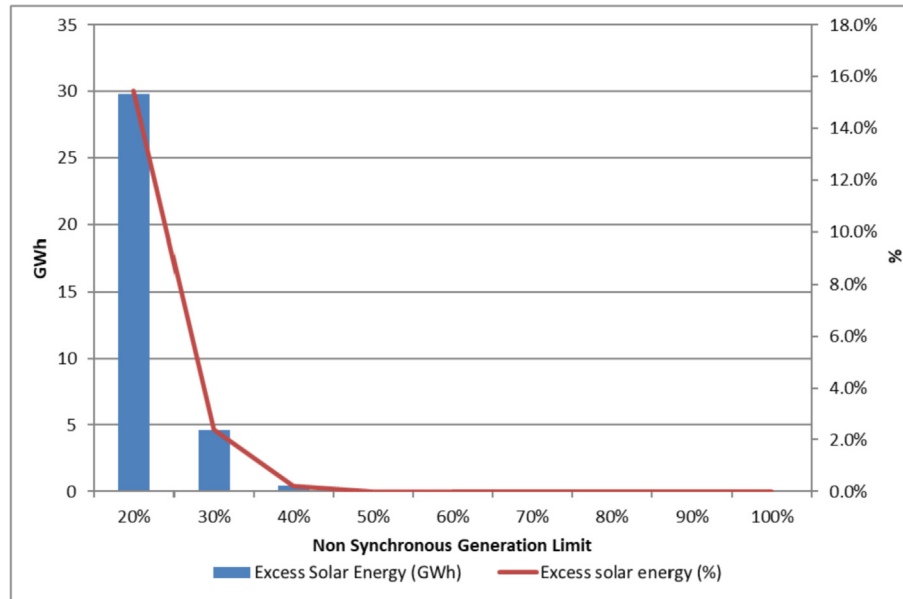


Fig. 5. Excess Solar Energy in 2020 with increasing NSG limit.

**Table 7**

Power plant output ramps under different NSG limits.

| NSG Limit | Max Ramp up (MW) | Max Ramp down (MW) | Number of hours with ramp ups |                  |                | Number of hours with ramp downs |                  |                |
|-----------|------------------|--------------------|-------------------------------|------------------|----------------|---------------------------------|------------------|----------------|
|           |                  |                    | >200 MW                       | >100 & = <200 MW | ≥0 & = <100 MW | >200 MW                         | >100 & = <200 MW | >0 & = <100 MW |
| 30%       | 86.5             | 103.0              | 0                             | 0                | 4243           | 0                               | 1                | 4539           |
| 40%       | 86.5             | 103.0              | 0                             | 0                | 4256           | 0                               | 1                | 4526           |
| 50%       | 86.5             | 103.0              | 0                             | 0                | 4266           | 0                               | 1                | 4516           |

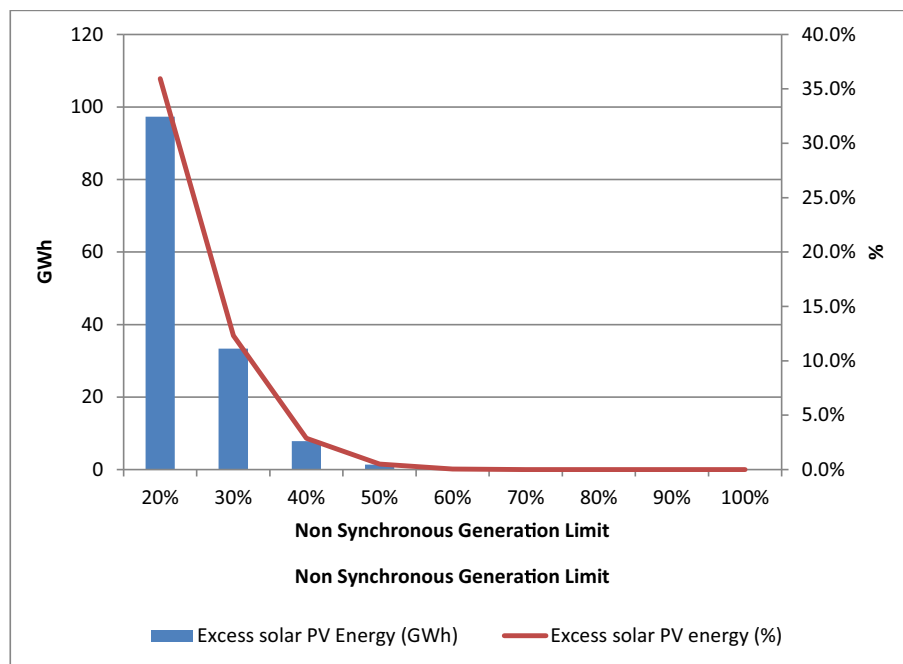


Fig. 6. Excess PV Energy with increasing NSG limit for LTES Scenario.

thought to be within the capabilities of the present Mauritan power system especially with the upgrades being performed.



**Table 8**  
Ramping requirements for LTES Scenario under different NSG limits.

| NSG Limit | Max Ramp up (MW) | Max Ramp down (MW) | Number of hours with Ramp ups |                  |                | Number of hours with ramp downs |                  |                |
|-----------|------------------|--------------------|-------------------------------|------------------|----------------|---------------------------------|------------------|----------------|
|           |                  |                    | >200 MW                       | >100 & = <200 MW | ≥0 & = <100 MW | >200 MW                         | >100 & = <200 MW | >0 & = <100 MW |
| 30%       | 114              | 121                | 0                             | 1                | 4155           | 0                               | 2                | 4625           |
| 40%       | 121              | 121                | 0                             | 6                | 4136           | 0                               | 5                | 4636           |
| 50%       | 121              | 121                | 0                             | 8                | 4142           | 0                               | 8                | 4625           |

As for the ramping requirements on the power plants, it can be seen in Table 7 that the highest ramp up event for the whole year is 86.5 MW and the ramp down is a maximum of 103 MW. Virtually all the hourly ramps are less than 100 MW.

Finally, the annual penetration of RE for year 2020 for the varying NSG limits was determined. Based on the committed projects, the RE penetration for 2020 is expected to be 25.4% with a minimum NSG of 40% (25.2% for an NSG limit of 30%).

### 3.3. Scenarios for 2025

#### 3.3.1. LTES scenario

As shown in Fig. 6, enabling operation of the Mauritian power system at an NSG limit of 50% enables almost all the solar energy to be integrated with only 1.4 GWh (0.4%) of excess electricity. This is to be contrasted with the situation of 2020 where an NSG limit of 40% was adequate.

Thus, the authorities must plan for a system which can operate at least with an NSG of 50% as this is the base scenario.

Due to the increased PV capacity in this scenario and the increased demand, the ramping requirements will increase accordingly. Hence, the maximum ramp up/down increase by 34.5/18 MW compared to the 2020 scenario as shown in Table 8. However, the number of larger ramps (>100 MW) is not expected to increase significantly. Finally, with an NSG limit of 50%, RE penetration in the system is 34.8% as shown in Table 9.

#### 3.3.2. PV scenario

As shown in Fig. 7, under this scenario, integration of more than 99% of the available solar PV energy necessitates that the power system be able to operate at an NSG limit of at least 80% which appears impossible in the medium term. At NSG limits of 70% the percentage/magnitude of excess PV is 2.6%/13.7 GWh and at NSG of 60% the excess generation rises to 6.0%/32.1 GWh which is unacceptably high for a country like Mauritius.

The massive increase of installed PV capacity in this scenario also reflects in the significantly greater ramping duty imposed on the power plants as shown in Table 10. Thus, the maximum hourly ramp up/down increases from 86.5 MW/103 MW in 2020 to 237 MW/222 MW in this scenario. The number of ramps also increases substantially with ramp ups greater than 100 MW for the case of NSG = 80% increasing from 0 in 2020 to 178 in this scenario.

Finally, the penetration of RE with increasing NSG levels is shown in Table 11 below. As it can be seen, at NSG limit of 70–80%, the share of solar PV is around 16.6% of the total electricity generation and this allows the target of 35% of RE to be reached for the electricity mix.

**Table 9**  
RE penetration with different NSG limits in LTES Scenario.

| NSG Limit | Solar (GWh) | % Solar in Electricity Mix | Total RE (GWh) | % RE  |
|-----------|-------------|----------------------------|----------------|-------|
| 30%       | 237.4       | 7.5%                       | 1067.7         | 33.8% |
| 40%       | 262.9       | 8.3%                       | 1093.3         | 34.6% |
| 50%       | 269.3       | 8.5%                       | 1099.7         | 34.8% |

Whilst the PV-dominated scenario might seem unrealistic, it certainly anticipates the system integration issues post-2025. Indeed, when all biomass and waste resources have been tapped using conventional generation there will come a point where further increase to the RE penetration must by necessity come from PV and/or wind generation. Apart from the issues of frequency stability which is well-known locally, operation at such high NSG levels would pose problem due to the minimum generation level of the power plants and the greatly increased ramping (load following) requirements, two essential issues which have, up to now, not been discussed in the local literature [4,11,15,28]. Fig. 8 depicts the issues of minimum generation level and high ramp requirements.

Additionally, the fact that the decrease in solar generation towards the afternoon correlates with the increase in demand for the evening peak creates a very challenging situation in terms conventional generation ramping requirement (MW/min) to meet the net load.

The issue of minimum generation limit can have important implications on the generation expansion strategy to be adopted by the utilities in island power systems. For example, prime movers having lower technical minimum, ability to start stop quickly and frequently, and better efficiency at part-load operation may be favored. Furthermore, instead of favoring power plants with larger individual units, it could be advantageous to opt for a larger number of smaller units which ultimately would lower the minimum overall generation of the whole power plant. The smaller unit size would also permit operation at better efficiency thus saving on fuel cost. In consideration of these factors, the choice of the conventional generation technology could be impacted. EPRI, for example, compared the challenges and opportunities of reciprocating internal combustion engine over OCGTs for peaking applications and ancillary services to counterbalance intermittent renewables [52]. It is to be noted that as the power system transitions to lower levels of synchronous generation, the system inertia and short-circuit level will drop. To alleviate these issues, it could be advantageous that future conventional generation be specified, where practicable, with the capability to function as synchronous condensers during periods of high renewables penetration and low residual demand. In fact, this technological solution has been instrumental in achieving up to 80% instantaneous penetration of solar PV (some of which backed by storage) in the island of Kauai in Hawaii'i [53].

The need for increased load-following/ramping capacity can also dictate the type of generation technology to be adopted. For example, increasing the generation capacity with base load plants such as slow responding steam turbines will not add the ramping capacity necessary to accommodate large share of VRE.

It is to be noted that for the latest proposed capacity addition the World Bank recommended  $2 \times 36$  MW OCGT based on lowest annual capacity utilization costs. The OCGTs are to form part of the gas turbine phase of a combined cycle power plant, if and when the supply of natural gas becomes a feasible option for Mauritius [26]. It appears that the issue of power system flexibility for facilitating renewables integration was not explicitly taken into consideration

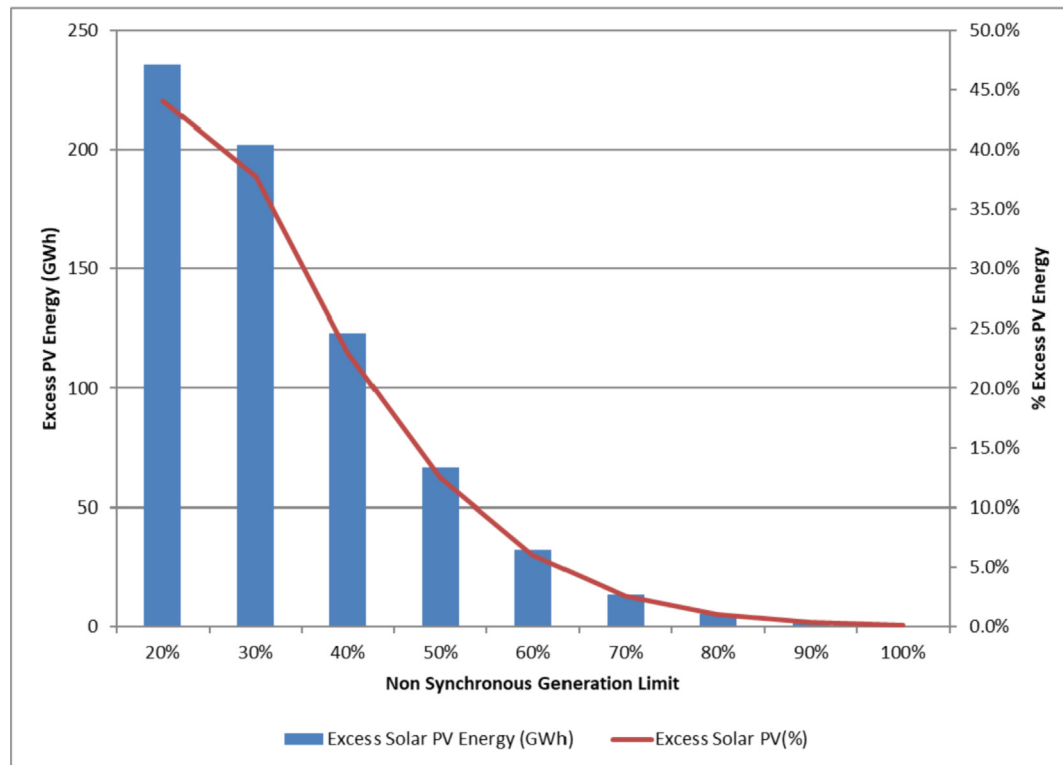


Fig. 7. Excess PV Energy with increasing NSG limit for PV Scenario.

Table 10

Ramping requirements for PV Scenario 2025 under different NSG limits.

| NSG Limit | Max Ramp up (MW) | Max Ramp down (MW) | Number of hours with ramp ups |                  |                | Number of hours with ramp downs |                  |                |
|-----------|------------------|--------------------|-------------------------------|------------------|----------------|---------------------------------|------------------|----------------|
|           |                  |                    | >200 MW                       | >100 & = <200 MW | ≥0 & = <100 MW | >200 MW                         | >100 & = <200 MW | >0 & = <100 MW |
| 40%       | 187              | 179                | 0                             | 53               | 3981           | 0                               | 26               | 4723           |
| 50%       | 223              | 185                | 1                             | 100              | 3870           | 0                               | 62               | 4750           |
| 60%       | 231              | 195                | 2                             | 136              | 3823           | 0                               | 94               | 4728           |
| 70%       | 237              | 222                | 4                             | 162              | 3831           | 4                               | 110              | 4672           |
| 80%       | 237              | 222                | 4                             | 174              | 3829           | 6                               | 128              | 4642           |

Table 11

RE penetration with different NSG limits in PV Scenario.

| NSG Limit | Solar (GWh) | % Solar in Electricity Mix (GWh) | Total RE (GWh) | % RE in Mix |
|-----------|-------------|----------------------------------|----------------|-------------|
| 40%       | 411.8       | 13.0                             | 990.4          | 31.3%       |
| 50%       | 468.0       | 14.8                             | 1046.5         | 33.1%       |
| 60%       | 502.6       | 15.9                             | 1081.2         | 34.2%       |
| 70%       | 521.1       | 16.5                             | 1099.6         | 34.8%       |
| 80%       | 529.5       | 16.7                             | 1108.0         | 35.0%       |

in the World Bank study which was the basis for the recommendation of the OCGTs.<sup>7</sup> For information, the gas turbine type earmarked for the project is the heavy duty/industrial gas turbine which is expected to have a minimum operating load of 50% [48].

### 3.3.3. PV plus Storage Scenario

It can be seen in Fig. 9 that the BESS reduces the amount of

excess solar energy. The 4-h BESS is more effective and a 2-h storage is insufficient for the PV installed capacity. A 140 MW BESS with 4-h storage capacity reduces curtailment from a maximum of 12.5%–2.8% that is it allows an additional 52 GWh of renewable energy to be integrated. This also permits the penetration of RE to reach a figure of 34.7% which is very near the target. However, as the BESS power capacity exceeds around 100–120 MW, the return decreases for increasing capacity.

To assess whether the use of BESS for time shift of solar energy will be cost-effective, the Levelized Cost of Storage (LCOS) was calculated using the following [54].

<sup>7</sup> The World Bank team emphasized that ideally, more formal expansion-planning and decision-making methodologies should be used, in line with discussions in the report, but this was not possible at that point in time due to the urgent need for new generation capacity.

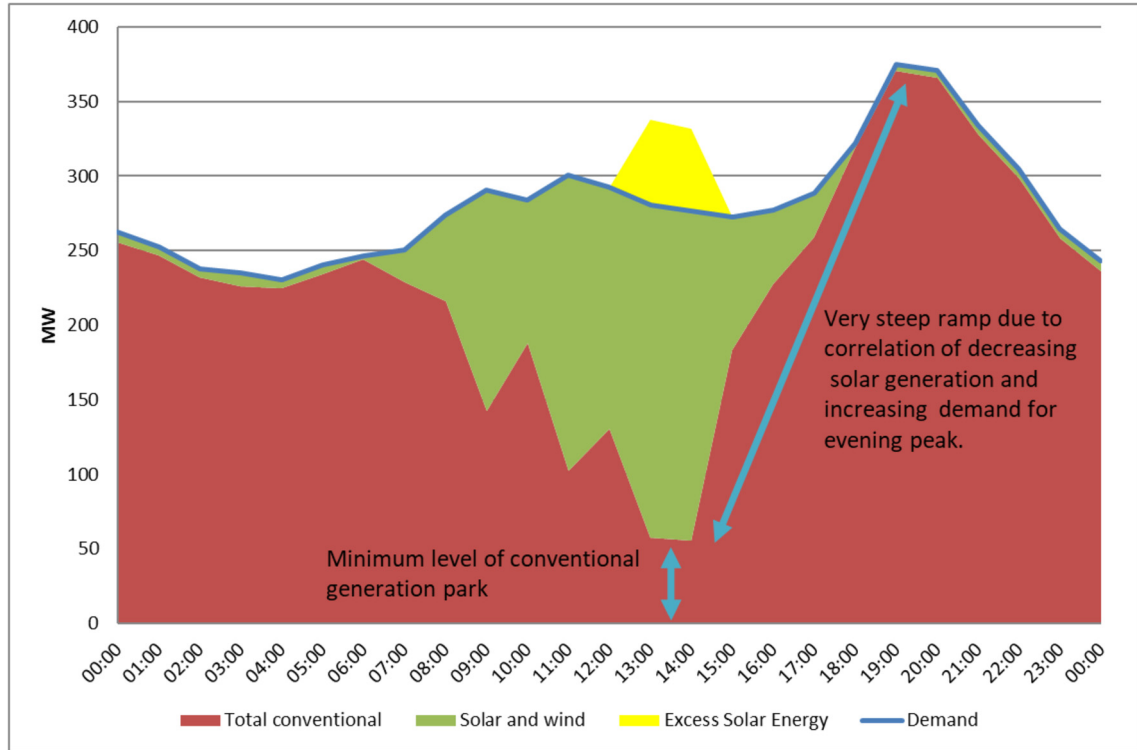


Fig. 8. Constraints posed by minimum generation limit and net load variations (NSG limit = 80%).

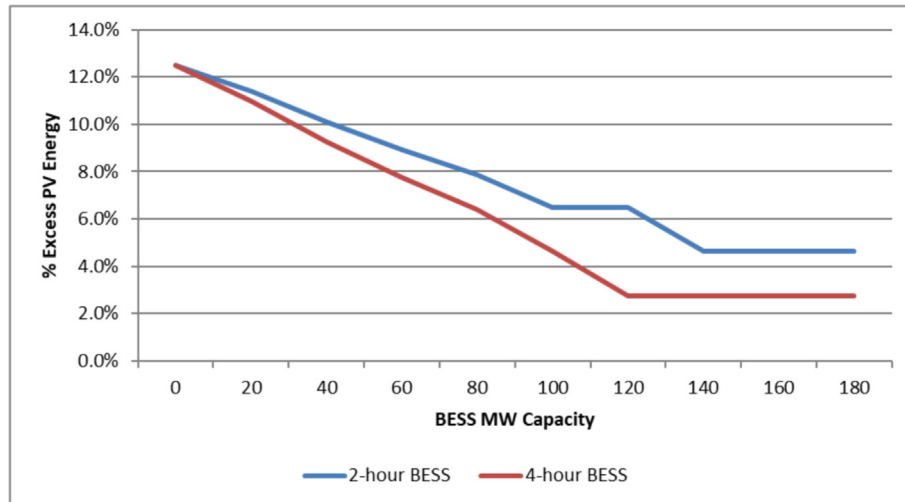


Fig. 9. Impact of BESS capacity on Excess PV Energy.

$$LCOS = \frac{\sum_{t=0}^n \frac{C_{BESS,t}}{(1+r)^t}}{\sum_{t=0}^n \frac{E_{out,t}}{(1+r)^t}} \quad (3)$$

where  $C_{BESS,t}$  is the total cost of the BESS at time  $t$ ,  $r$  is the discount rate assumed to be 6%,  $n$  is the total duration the BESS is expected to be in operation, i.e. 15 years and  $E_{out,t}$  is the annual solar energy delivered by the BESS. The installation and maintenance costs of the BESS for year 2025 is assumed as in reference case of [55]. The

results are shown in Fig. 10. The LCOS for the 2-h system is lower than the 4-h system for all system sizes which indicate that the amount of solar energy that it captures is proportionally higher than the 4-h system. However, as noted above the 2-h BESS is not enough to reduce excess solar energy to acceptable level. For the 4-h BESS, the lowest LCOS is of the order of USD 0.37/kWh (MUR 13/kWh). This is a very high figure compared to the average sales price of electricity which is USD 0.16/kWh (MUR 5.73/kWh) [2].

Analyzing further the performance of the BESS reveals that the reduction of annual excess PV energy is consistently exactly half the BESS annual electricity input. This peculiarity is due to the

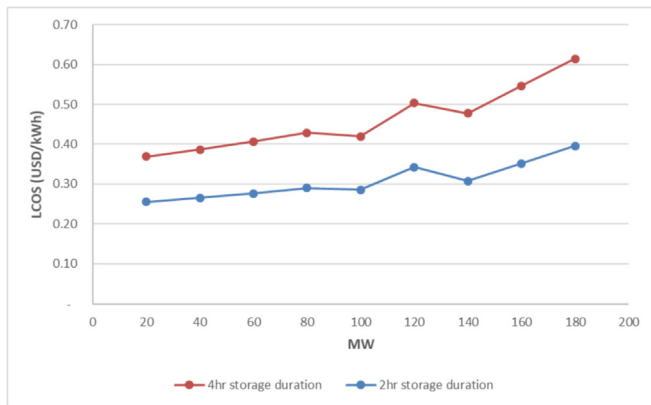


Fig. 10. LCOS for varying BESS capacity.

constraint of operating the power system at an NSG limit of 50%. Consequently, when the BESS is charging, the amount of additional instantaneous solar generation that it can enable on the network is limited to only 50% of the BESS power capacity otherwise the NSG limit would be violated. Thus, the BESS is under-utilized in regards of solar PV integration. Based on the foregoing assumptions and results, it can be concluded that the use of alternating current (AC)-coupled BESS for large-scale energy-time shift (curtailment reduction) of solar PV is not a viable option for Mauritius up to 2025.

The results suggest that to circumvent this issue, it would be required to be able to charge the BESS with PV energy without the PV energy transiting through the synchronous grid.<sup>8</sup> In effect, it would be necessary to couple the PV generator and the batteries on the direct current (DC)-side instead of the AC-side. The DC-coupled PV-plus-storage can then be charged exclusively with solar energy that cannot be accepted by the synchronous grid, greatly reducing curtailment, improving the LCOS of the BESS while at the same time not causing violation of the NSG limit. The use of DC-coupled solar plus storage would also allow using very high inverter loading (DC:AC ratio) without causing clipping which will greatly increase the capacity factor of the solar PV plants and give a boost to the annual penetration of RE in the system. This might be an interesting solution for island power systems which are nearing saturation.<sup>9</sup> Reference [56] provides an overview of the pros and cons of this relatively new technology. DC-coupled solution can nearly halve losses compared to the AC variant and enable larger energy export in grid constrained projects [57].

### 3.3.4. Curtailment scenario

As it can be seen in Fig. 11, curtailment (approx. 0.1%) starts to be experienced when the amount of solar PV energy reaches around 230 GWh which corresponds to around 7.3% of the annual demand. The corresponding PV installed capacity is 140 MW<sub>ac</sub>.

The yearly penetration of PV energy can be greatly increased by

allowing a greater curtailment rate. For example, assume that the maximum curtailment rate is arbitrarily set to 2%. This results in a yearly PV penetration of approx. 340 GWh (installed capacity of 210 MW) i.e. an increase of 110 GWh which is 47.8% over the case of no allowable curtailment and represents an annual penetration of 10.8%. However, in case of curtailment, the local grid operator could still have to pay the plant owner for the curtailed energy depending on the terms of the Energy Supply and Purchase Agreement. Whilst percentage wise the figures may appear negligible; in absolute terms the cost of curtailment may become very high. However, from a pragmatic point of view, if the share of VRE in the electricity mix is to increase further in a cost-effective manner, there must be a point where some modest amount of curtailment must be allowed.

## 4. Conclusions

There was an important research gap concerning the adequate modelling of the Mauritian island energy system which hampered the investigation of how to effectively transition to a sustainable system. Recent studies focused on longer-term policy issues and did not explore technical challenges that must be tackled in priority. In that regards, this study has laid the foundation stone by creating a model for the entire energy system for year 2016 and which is geared towards the study of large-scale integration of VRE. The model has been validated by comparing its output with that of statistics of year 2016. The differences between the simulation and reality were found to be small enough so that the model be considered accurate. With that goal fulfilled, attention has been turned to the LTES. The anticipated system for 2020 has been simulated taking into consideration the capacity additions of both committed VRE and conventional plants. The results show that the system can integrate the anticipated increase in VRE without difficulty and the penetration of RE is expected to grow to around 25.4%.

The situation for 2025 was next analyzed with the help of four scenarios to capture the whole range of possibilities and consequently provide useful insight as well as important policy implications. As expected, attaining the LTES target through exploitation of bagasse is the least demanding option in terms of transformation of the power system in that the conventional generation fleet needs to be able to sustain operation at a maximum NSG of 50%. However, given the uncertainty associated with the LTES pathway and to anticipate the technical challenges, three other scenarios were modelled. In a PV-dominated scenario, the system must be able to operate at an NSG of at least 80% to keep the curtailment rate to less than 1% and achieve the LTES target. Even though this performance appears impossible to achieve by 2025, there are a number of important insights as follows: regardless of the pathway adopted, the conventional generation fleet must be made more flexible and to achieve this, the current generation expansion methodology<sup>10</sup> as detailed in Ref. [15] must be adapted to incorporate the key requirements of flexibility. This is especially important given the huge investment and long lifetime of the generation plants. A suboptimal expansion methodology might slow the transition to a sustainable energy system.

Under the realistic constraint of an NSG limit of 50%, the use of traditional AC-coupled BESS for energy-time shift (curtailment recapture) of excess PV generation was found to be economically unviable due to low utilization. To circumvent the NSG constraint or any grid stability-related constraint that limits the injection of VRE

<sup>8</sup> In fact, as per [28], the upgraded power system of Mauritius will have a VRE grid absorption capacity of 185 MW. This is not enough for reaching the LTES target in the PV Scenario. While it is possible to install the BESS on the AC side, and charge it with excess PV energy exported to the grid, this configuration appears to be more challenging for system operation as it is likely that the effects of instantaneous PV output fluctuations would still be felt on the grid with serious implications on frequency stability. The round-trip efficiency would also be less advantageous.

<sup>9</sup> As at today, the committed capacity for VRE is around 154.75 MW while the grid absorption capacity of the upgraded Mauritian power system is 185 MW. Clearly, with a few more utility-scale projects, the limit will be reached. Currently, there appear to be no published strategy detailing how the grid might be transformed to increase the absorption capacity further.

<sup>10</sup> The choice of generation type is based on analysis of the evolution of the Load Duration Curve and using a Screening Curve.

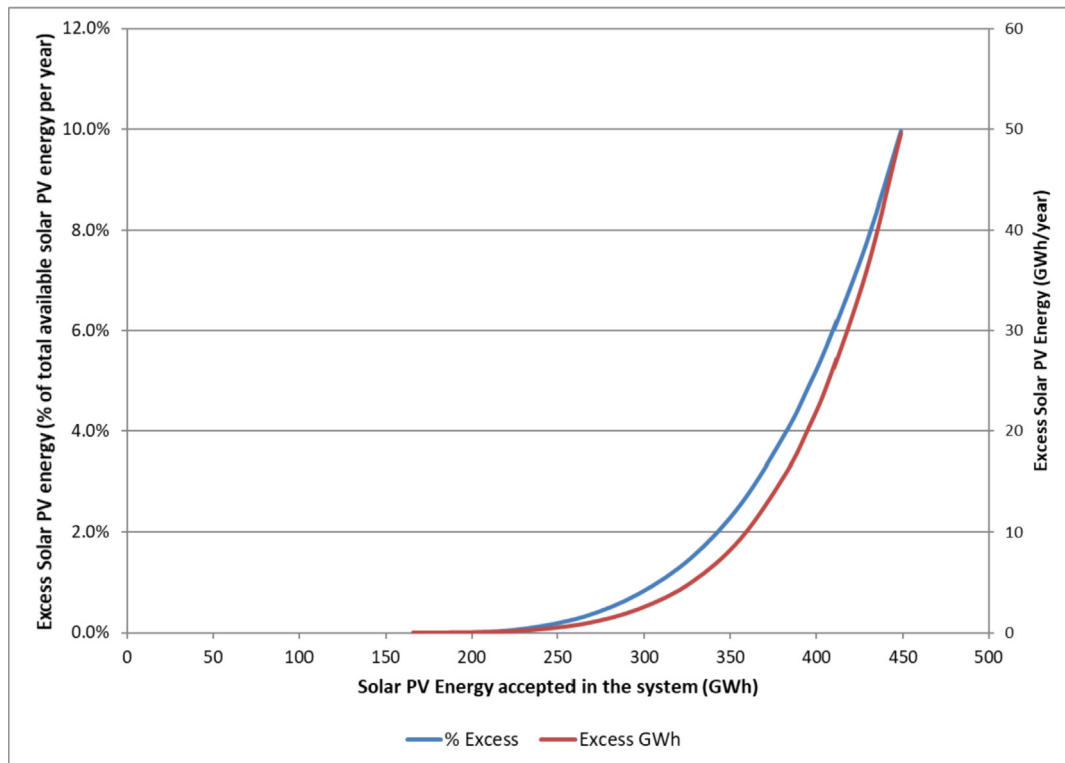


Fig. 11. Variation of excess PV energy with penetration.

into the grid, it is suggested to investigate the application of *DC-coupled Solar plus Storage*. Future research will investigate the technical and economic feasibility of this solution for Mauritius.

It was also found that allowing for a modest amount of curtailment could greatly increase the solar energy penetration. It remains an important research question to determine what the optimal level of curtailment would be in Mauritius. The issue of excess solar energy needs also to be examined from the perspective of future wide scale adoption of electric vehicles in Mauritius, as the MARENA calls for a policy to phase out fossil-fuel based vehicle by 2040 [25]. Through a smart charging system, it may be possible to simultaneously increase the penetration of RE in the electricity system and decarbonize the transport sector.

Whilst this work concentrated on the average curtailment rate, it is important that further research be performed to determine the marginal curtailment rate which can indicate the threshold at which PV becomes uncompetitive with alternative sources [32]. Further research must also be performed on the effectiveness of Demand Side Management to increase system flexibility.

It is hoped that the results of this study can foster fruitful discussions amongst planners and policy makers in Mauritius with the objective of improving the planning for large scale integration of VRE in the island. In fact, this work explored future technical challenges and potential solutions that have not been discussed in the latest official strategy [11]. This study should also be of interest to the authorities of many small island developing states who are facing similar constraints concerning the development of renewable energy in their territories.

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